

Parallax Effect

In simple terms, a Parallax is when the distant objects seem to “move” slower than the closer objects.

Example: You're riding a car. You see buildings along the road. And you see mountains far away.

As you travel, the buildings that are closer to you, are passing by faster.

While mountains, that are far away from you, are passing by slower.

The Parallax effect gives animations a dimension and depth.

The Mountains
They're passing by slower

The Buildings
They're passing by faster



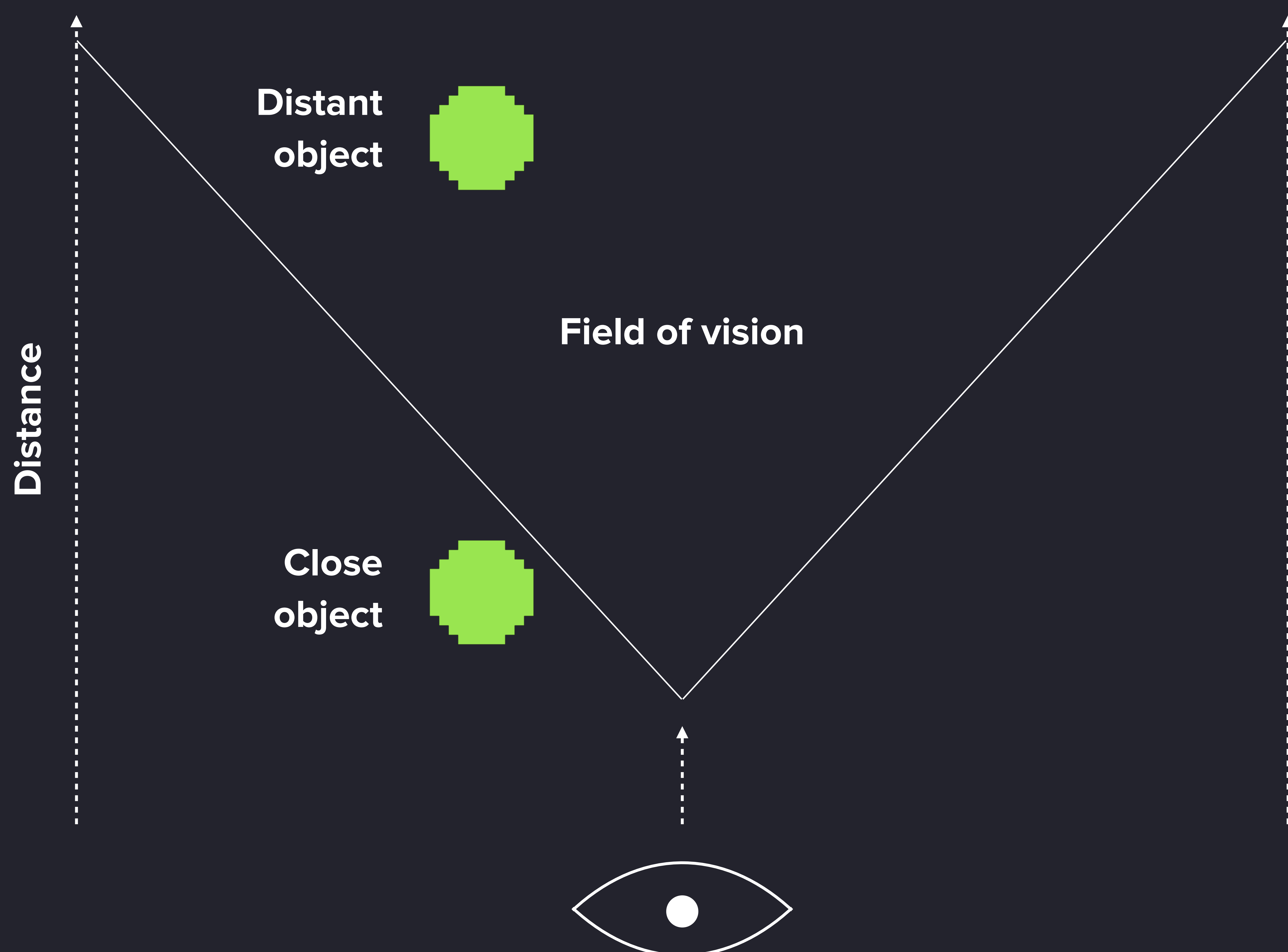
Optical illusion behind the Parallax Effect

In fact both, the distant and close objects “move” with the same speed.

However, the field of our vision expands with distance.

And it takes distant objects longer to travel through our field of vision, then the closer objects.

That’s why, it seems like the distant objects move slower.



Parallax Scrolling

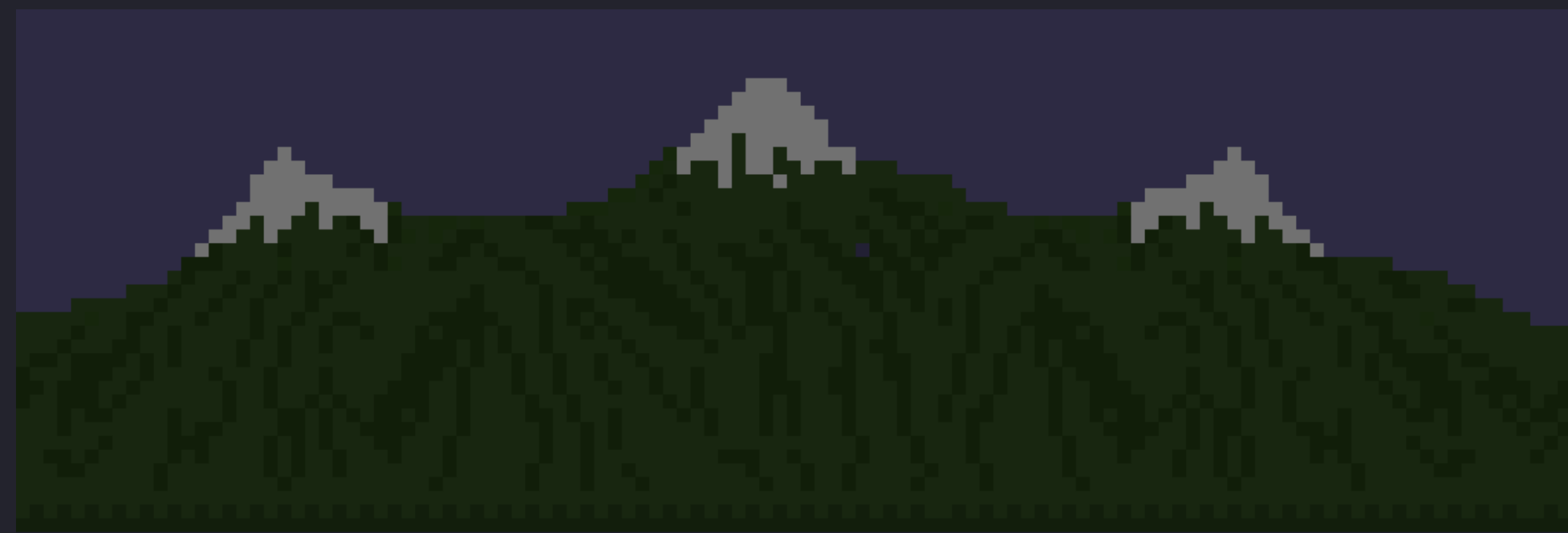
Parallax scrolling is made of several layers that move with different numbers of pixels at a time.

The closer a layer to a viewer, the more pixels it moves at a time.

The more distant a layer is to a viewer, the fewer pixels it moves at a time.

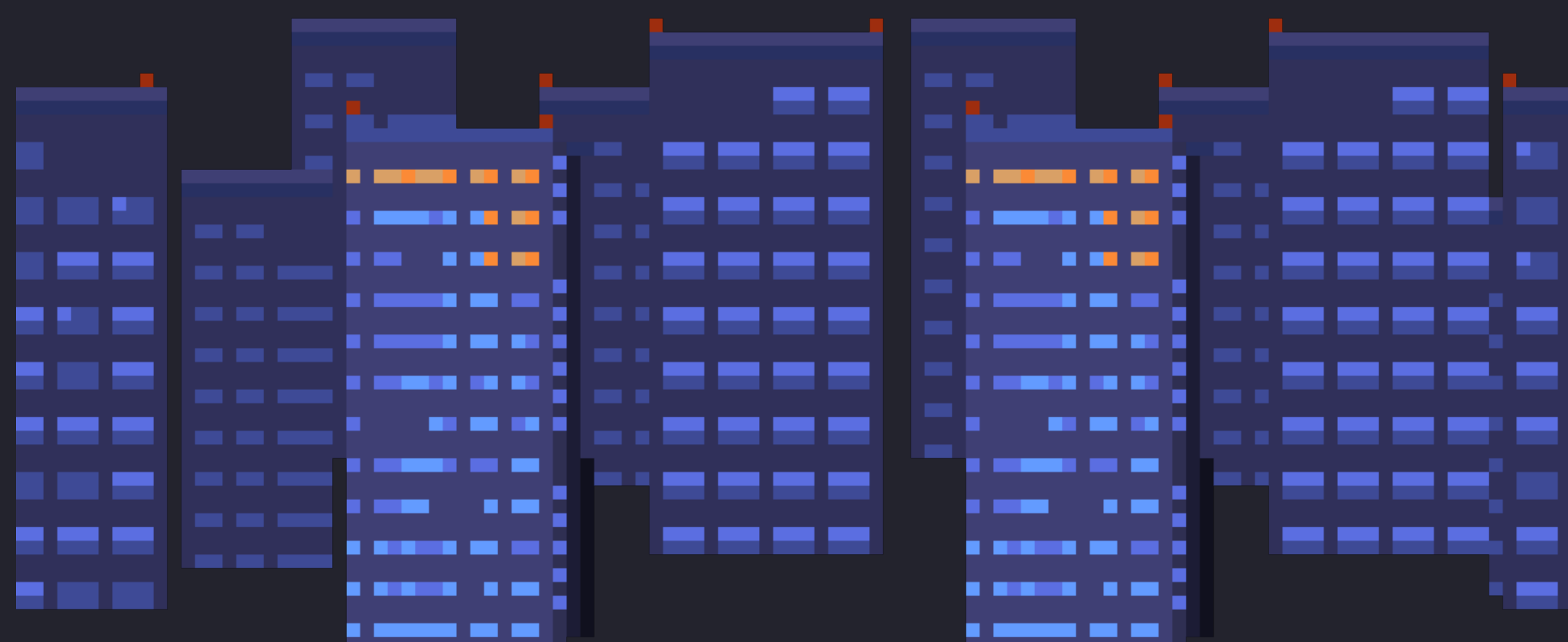
The layers are usually organized into Foreground, Midground, and the Background.

Layer 4
Moves 1 px at a time



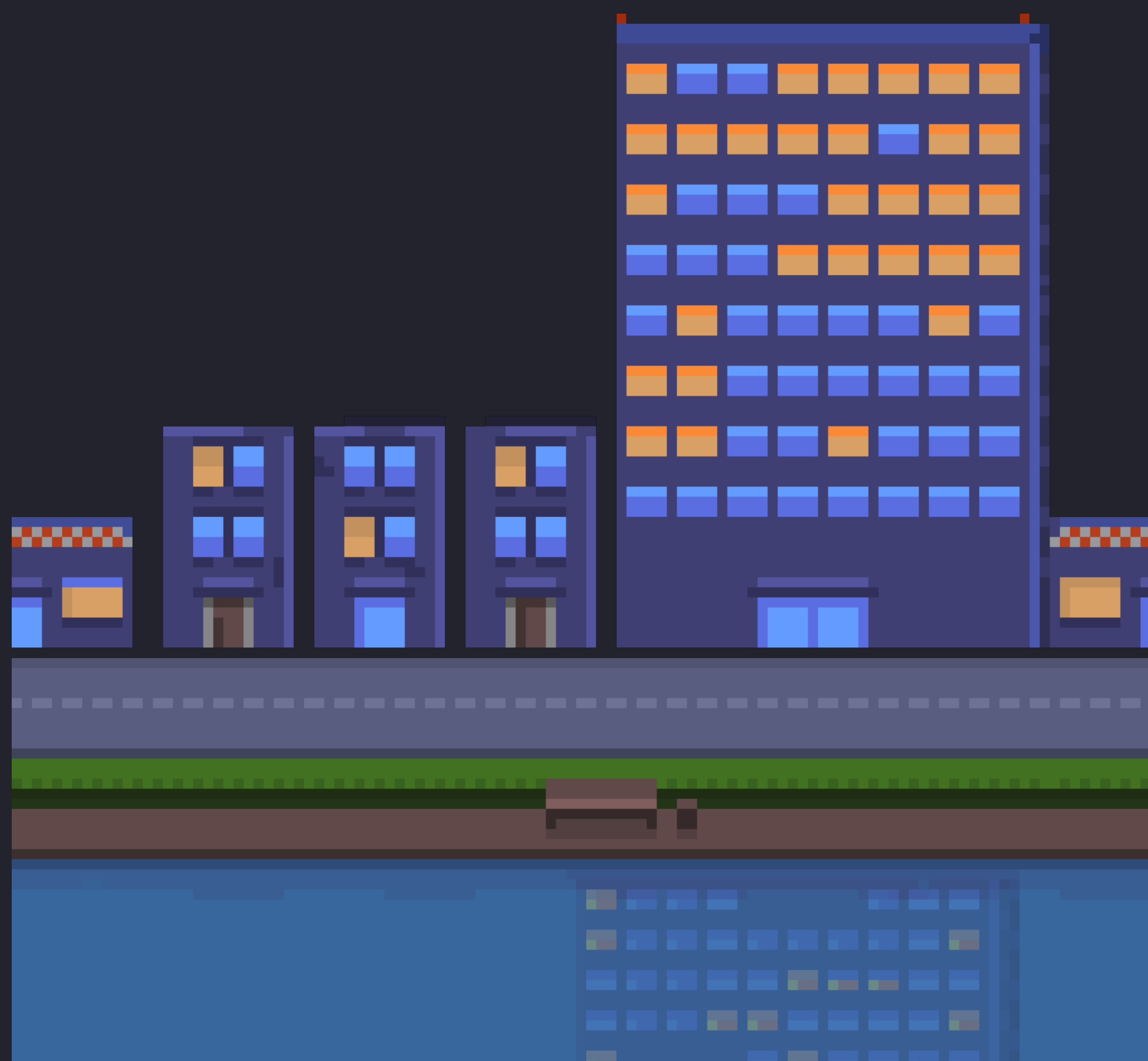
Background

Layer 3
Moves 2 px at a time



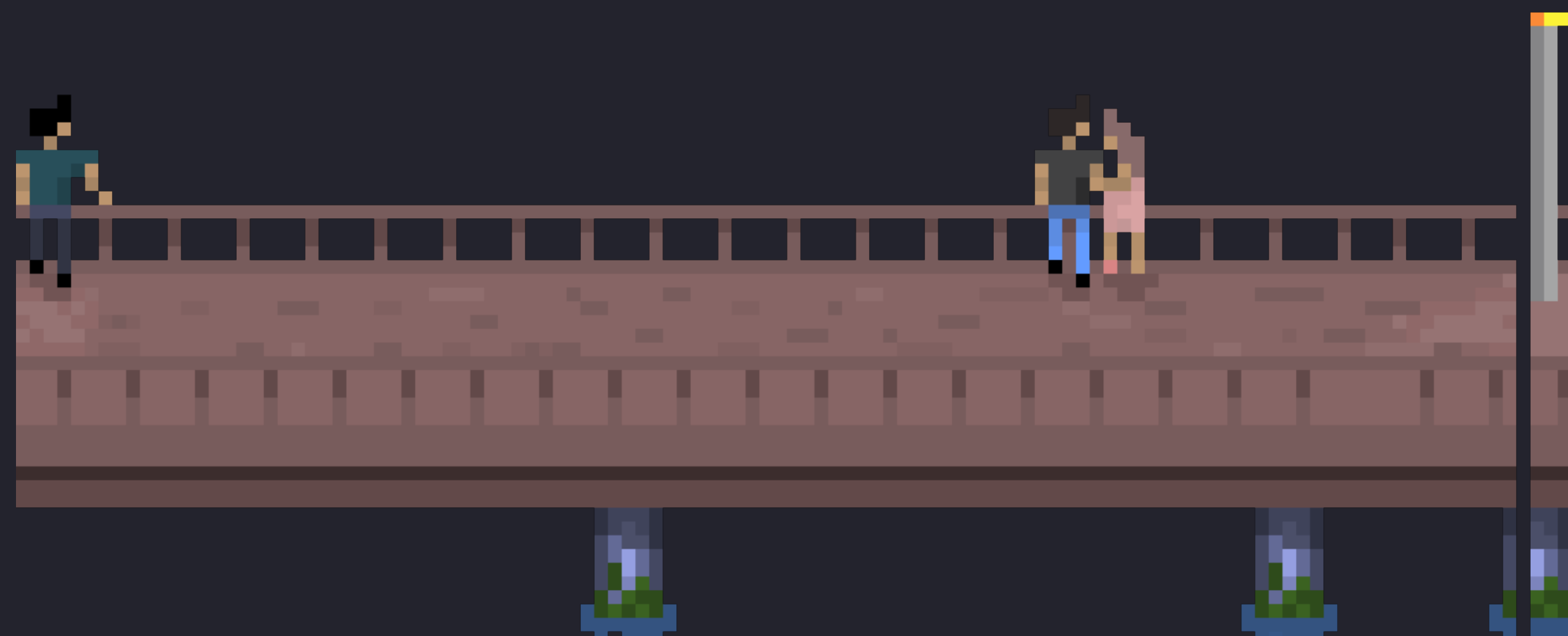
Midground

Layer 2
Moves 3 px at a time



Foreground

Layer 1
Moves 4 px at a time



Parallax Effect

To create a Parallax loop, the number of parallax frames should correspond to the canvas width (for horizontal scrolling) and height (for vertical scrolling)



114 px canvas = 114 Parallax animation frames